

Assignment 1: Shared Task / Challenge Project

SemEval 2027 / ICASSP 2027

Purpose: understand the task, data, evaluation setup, and establish reproducible baselines before advanced modeling.

Expected progression: Task understanding -> Data understanding -> Selective EDA -> Classical ML baseline -> Pretrained model baseline -> Evaluation -> Error analysis -> Advanced system

1. Task Understanding

Begin with a black-box understanding of the shared task, then move to a white-box understanding of what linguistic, semantic, acoustic, or dataset properties may determine the target.

- Identify the input, output, prediction type, label space, and official evaluation metric.
- Clarify whether the task is classification, regression, ranking, retrieval, generation, detection, segmentation, or another formulation.
- Inspect the task description without modeling assumptions first: What is observable? What must be predicted?
- Then move to white-box reasoning: Which linguistic/acoustic cues could be relevant? Which confounds, shortcuts, annotation artifacts, imbalance, domain, speaker, or source effects may exist?
- Read the shared-task description, organizer baseline/system paper, and closely related literature.

Deliverable: a concise task formulation covering input, output, labels/targets, metric(s), and a short explanation of what makes the task difficult.

2. Data Understanding and Selective EDA

Important: You are NOT expected to compute, discuss, and interpret every possible feature. EDA must be selective and task-driven. Choose a small, justified set of features that helps you understand the dataset and the prediction problem.

2.1 Minimum dataset-level checks

- Number of samples and train/dev/test split sizes
- Label/target distribution and imbalance
- Missing, empty, duplicated, or unusually short/long samples
- Basic length statistics: tokens/characters for text; duration for speech
- Relevant metadata distributions: source, language, speaker, domain, recording condition, etc.

2.2 Feature selection rule

Select approximately 4-8 informative feature groups in addition to the basic dataset checks. The exact number depends on the task. Every selected feature should answer a question about the task, dataset, or likely model behavior.

Examples for text/NLP tasks

Feature group	Possible measures	Useful libraries
Length / structure	token count, sentence count, average sentence length	NLTK, spaCy, pandas
Lexical richness	TTR, MSTTR, MTLD, HD-D, hapax proportion	lexical-diversity, NLTK

Feature group	Possible measures	Useful libraries
Readability / complexity	Flesch, Flesch-Kincaid, Gunning Fog, SMOG	textstat
Lexical usage	word/character n-grams, word frequency, stop-word ratio	scikit-learn, NLTK, wordfreq
Morphosyntax	POS proportions, dependency patterns	spaCy, stanza
Semantics	sentence embeddings, semantic similarity, cluster structure	sentence-transformers, transformers
Task-specific cues	sentiment, emotion, negation, entities, modality, domain terms	spaCy, NLTK, task-specific libraries

Examples for speech/audio tasks

Feature group	Possible measures	Useful libraries
Signal / duration	duration, silence ratio, RMS energy, zero-crossing rate	librosa, torchaudio
Prosody	F0/pitch, pitch range/variation, speaking rate	praat-parselmouth, openSMILE
Spectral	spectral centroid/bandwidth/roll-off, MFCCs	librosa, openSMILE
Time-frequency	Mel-spectrograms, spectrogram summaries	librosa, torchaudio
Voice quality	jitter, shimmer, HNR where task-relevant	praat-parselmouth, openSMILE
Metadata	speaker, channel, device, language/accent, recording setting	pandas + dataset metadata

2.3 How to analyze and interpret EDA

For each selected feature, do not stop at a mean, histogram, or boxplot. Use the following three-step pattern:

1. Observation: state the visible/statistical pattern.
2. Interpretation: explain what the pattern may mean for the task or data.
3. Modeling implication: state a plausible consequence for feature engineering, baselines, evaluation, or error analysis.

Example A - Text length

Observation: Class A documents are substantially longer than Class B documents.

Interpretation: Length may encode class-related structure, but it may also represent a dataset shortcut rather than the intended phenomenon.

Modeling implication: Include length in a classical baseline, but also test whether performance drops when length is controlled or normalized.

Example B - Vocabulary richness

Observation: One label has higher MTLT/HD-D values while raw TTR decreases with document length.

Interpretation: The class may differ in lexical diversity; however, raw TTR is length-sensitive, so robust diversity measures are more reliable for comparison.

Modeling implication: Use one or two robust richness measures rather than many redundant variants.

Example C - Speech pitch/energy

Observation: A target class shows larger pitch variability but similar mean pitch.

Interpretation: Prosodic dynamics may be more informative than absolute F0 level; speaker identity or gender imbalance must be checked before attributing the pattern to the task label.

Modeling implication: Prefer normalized pitch-range/variation features and inspect results across speakers or metadata groups.

What NOT to write: “Class A has an average sentence length of 18.2 and Class B has 15.6.” This only reports numbers. Add what the difference may indicate, whether it is trustworthy, and why it matters for the task.

2.4 Visualizations and EDA PDF

- Use plots only when they reveal a relevant pattern: distributions by class, box/violin plots, correlation heatmaps, lexical/readability comparisons, embedding projections, spectrograms, pitch/energy distributions, etc.
- Do not create graphs merely because a library provides them. Every figure must support a question or observation.
- Save the selected EDA figures in a single PDF.
- Do not add traditional figure captions in the PDF. Instead, make the plot self-explanatory using a concise title, meaningful axis labels, units, and legend where needed.

Example plot labeling: Title: “Document Length Distribution by Class”; X-axis: “Number of Tokens”; Y-axis: “Frequency”.

3. Baseline Development

Baseline A - Classical ML with interpretable features

- Build a feature matrix from selected EDA/task-relevant features.
- For text, also establish TF-IDF + Logistic Regression and/or Linear SVM where appropriate.
- For speech, use MFCC/openSMILE/acoustic features with models such as SVM, Logistic Regression, Random Forest, or another justified classical learner.
- Inspect feature importance/coefficients where methodologically valid and connect them back to EDA.

Typical models: Logistic Regression | Linear SVM | Naive Bayes | Random Forest | XGBoost/LightGBM

Baseline B - Pretrained models

- Move to pretrained models only after establishing the classical baseline.
- Text examples: BERT, RoBERTa, DeBERTa, XLM-R, or a suitable domain/task-specific encoder.
- Speech examples: wav2vec 2.0, HuBERT, WavLM, Whisper representations, or task-specific speech encoders.
- State clearly whether you use frozen representations, feature extraction, fine-tuning, or prompting/instruction-based inference.

4. Evaluation and Literature Understanding

- Understand the official metric mathematically and conceptually.
- Explain why the shared task uses it and what behavior it rewards or penalizes.
- Identify macro/micro/weighted averaging where applicable.
- Discuss effects of class imbalance, skew, or label structure.

- Read at minimum: the shared-task description, organizer baseline/system paper, and 3-5 closely related papers.
- Focus literature reading on task formulation, data, features, models, evaluation setup, and limitations - not abstract summaries.

Possible metrics: Accuracy | Precision/Recall/F1 | Macro-F1 | Micro-F1 | Pearson/Spearman | MAE/RMSE | WER/CER | BLEU/ROUGE/BERTScore | task-specific metrics

5. Report Writing - LaTeX / Overleaf

All reports must be written in LaTeX using Overleaf. Use proper sectioning, BibTeX references, mathematical notation, and cross-references for tables/figures.

1. Introduction and Task Definition
2. Related Work
3. Dataset
4. Exploratory Data Analysis
5. Feature Engineering
6. Baseline Models
7. Evaluation Setup
8. Results
9. Error Analysis
10. Discussion
11. Next Steps

Useful LaTeX packages: graphicx | booktabs | amsmath | hyperref | cleveref (optional)

Assignment 1 Submission Checklist

- ☐ Clear black-box and white-box task understanding
- ☐ Dataset statistics and quality checks
- ☐ Justified selection of task-relevant EDA features
- ☐ Interpretation of selected features using Observation -> Interpretation -> Modeling implication
- ☐ Single PDF containing relevant EDA plots
- ☐ At least one classical ML baseline using engineered/task-relevant features
- ☐ At least one appropriate pretrained-model baseline
- ☐ Correct understanding of official evaluation metric(s)
- ☐ Initial literature review grounded in task/baseline papers and related work
- ☐ LaTeX/Overleaf report with reproducible experimental description

Core principle: A high leaderboard score is not sufficient. The project must demonstrate understanding of the task, data, linguistic/acoustic evidence, baseline behavior, evaluation metric, and failure cases.